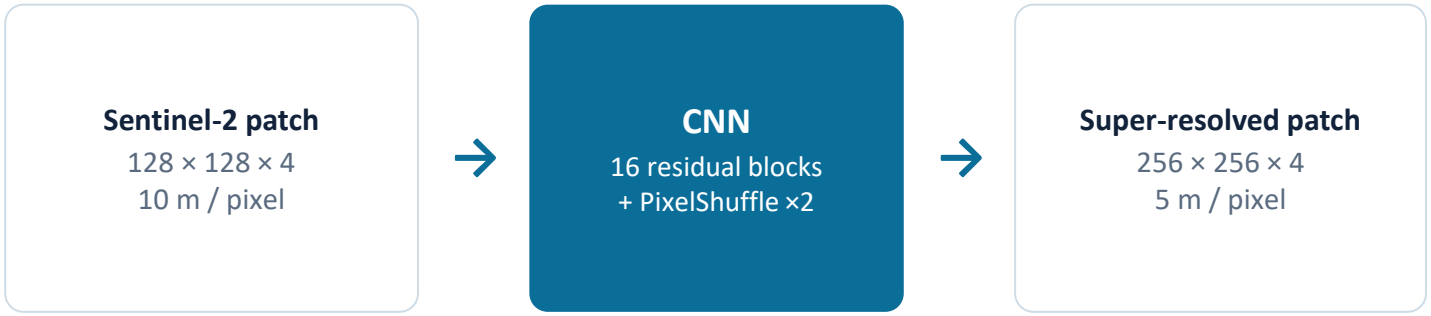
Super-resolution of Sentinel-2 satellite imagery

Reconstructing 5 m detail from 10 m imagery while preserving spectral fidelity | NNDL course project

PROBLEM

Sentinel-2 imagery is openly available at 10 m per pixel; commercial imagery at 1–2 m is costly. The objective is to train a CNN that reconstructs fine spatial detail from the open imagery while preserving the band values on which downstream analytics such as NDVI depend.

INPUT AND OUTPUT



- Bands: B2 blue, B3 green, B4 red, B8 near-infrared, 10 m native
- Upscaling factor ×2, 1.37 M parameters
- Loss: L1, with a spectral term evaluated as an ablation

DATA

SEN2VENµS

Sentinel-2 patches (10 m) paired with VENµS acquisitions (5 m) of the same location and date.

- Real acquisitions at both resolutions, not synthetic blur
- Zenodo 14603764, 29 sites, non-commercial licence
- 1,846 patches from ESTUAMAR and SUDOUE-4: 1,661 for training, 185 held out
- 129 patches from FGMANAUS, held out from training

Reflectance: int16 scaled by 10000, clipped to [0, 1].

Evaluation protocol

Each low-resolution patch has a corresponding high-resolution capture of the same location. All scores are computed against that reference.

PARTITIONING

Splits are defined by site. Patches within a site cover adjacent ground, so a patch-level split would place near-duplicates in both partitions. Ten percent of the patches from the training sites are held out as an in-domain test set, and a third site is held out entirely as an out-of-domain test set.

METRICS

Metric	Quantity	Better
M1 PSNR	Reconstruction error	**higher ↑**
M2 SSIM	Local structure, edges and texture	**higher ↑**
M3 SAM	Spectral angle distortion across bands	**lower ↓**
M4 NDVI MAE	Error in the derived vegetation index	**lower ↓**
M5 |D(SR)−LR|	Agreement with the input measurement	**lower ↓**

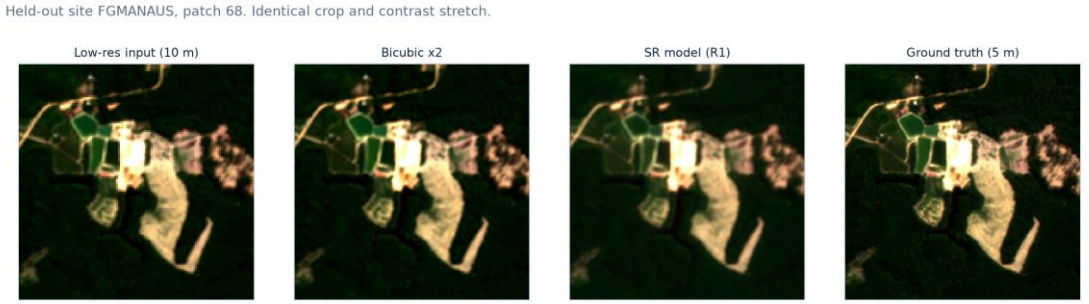
Bicubic interpolation introduces no high-frequency content, so its downsampled output closely reproduces the input. M5 therefore favours blurred predictions and must be interpreted alongside M1 and M2.

IN-DOMAIN TEST SET

Method	PSNR ↑	SSIM ↑	SAM ↓	NDVI ↓	M5 ↓
Bicubic	42.159	0.9378	1.582	0.02810	0.00078
CNN L1	43.774 ±0.038	0.9506 ±0.0003	1.295 ±0.008	0.02415 ±0.00013	0.00464 ±0.00002
CNN L1+λSAM	43.731 ±0.050	0.9500 ±0.0005	1.292 ±0.009	0.02413 ±0.00018	0.00460 ±0.00008

- The network exceeds bicubic by +1.61 dB PSNR and improves both spectral metrics.
- Bicubic interpolation is the standard non-learned reference. Out-of-domain results are on slide 5.

QUALITATIVE COMPARISON: IDENTICAL CROP AND STRETCH



Methodology and contribution

Why the proposed spectral loss term is redundant, and what is measured in its place

$$\mathrm{SAM}(a \cdot p,\ t) \;=\; \mathrm{SAM}(p,\ t) \qquad \text{for every } a > 0$$

SAM is invariant to scaling of the predicted vector, so its minimum coincides with the minimum of L1.

ANALYTICAL ARGUMENT

- L1 constrains every band of every pixel, which forces the spectral angle to zero without an additional term
- Both terms reach zero at the same parameter setting, so λ · SAM introduces no additional optimum
- The term reweights an existing error signal without adding a constraint. The same reasoning applies to an NDVI loss term

A1 · GRADIENT ALIGNMENT

$$\cos\left(\nabla L1,\ \nabla \mathrm{SAM}\right)$$

The two terms are differentiated separately every 100 steps and the angle between the resulting gradient vectors is recorded.

- cos = 1: the SAM gradient is parallel to the L1 gradient
- cos = 0: orthogonal, supplying information L1 does not
- cos < 0: the two gradients oppose each other

Gradient-conflict diagnostic from multi-task learning (Yu et al., NeurIPS 2020).

PLANNED EXTENSION

L1 compares the prediction with the target only. A back-projection term compares the prediction with the input ($\mu \cdot |D(SR) - LR|$) and therefore constrains a quantity that is absent from L1. In remote sensing this is the consistency property of Wald's protocol (1997).

Experiments and benchmarks

Each row differs from the preceding row in a single component

RUNS

Run	Method	Purpose	Status
R0	**Bicubic ×2**	Non-learned reference	complete
R1	**CNN, L1**	Baseline network	3 seeds
R2	**CNN, L1 + λ · SAM**	Effect of a spectral loss term	3 seeds
R3	CNN, L1 + μ · consistency	Effect of an observation constraint	specified
R4	R3 + uncertainty head	Calibration of predicted confidence	specified

REPLICATION

R2 is expected to produce only a small change. Differences below the run-to-run standard deviation cannot be separated from initialisation noise, so R1 and R2 are each run with three seeds.

CONTROLS

- λ fixed at 0.1, selected on in-domain results

BENCHMARK

- Bicubic interpolation is the standard non-learned reference
- Absolute PSNR depends on the dataset and band selection; the controlled difference between rows is the quantity of interest

EVALUATION REGIMES

Each run is scored on an in-domain test set and on the held-out site FGMANAUS. The first indicates whether the model performs the task; the second indicates whether performance is retained under a change of land cover. The two results differ.

Results

Six training runs. Three seeds each for R1 and R2, scored on both evaluation regimes.

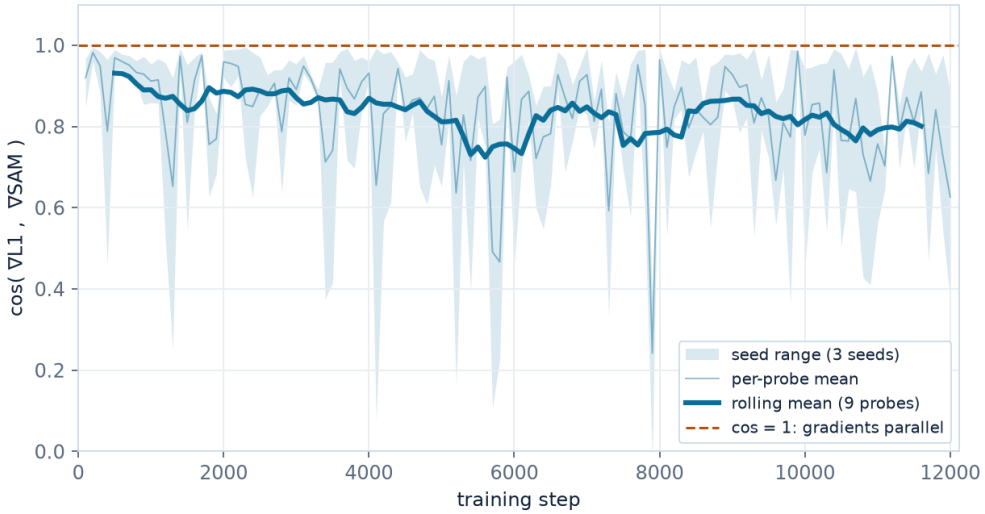
IN-DOMAIN TEST SET: HELD-OUT PATCHES, TRAINING SITES

Method	PSNR ↑	SSIM ↑	SAM ↓	NDVI ↓	M5 ↓
Bicubic	42.159	0.9378	1.582	0.02810	0.00078
CNN L1	43.774 ±0.038	0.9506 ±0.0003	1.295 ±0.008	0.02415 ±0.00013	0.00464 ±0.00002
CNN L1+λSAM	43.731 ±0.050	0.9500 ±0.0005	1.292 ±0.009	0.02413 ±0.00018	0.00460 ±0.00008

OUT-OF-DOMAIN: HELD-OUT SITE FGMANAUS

Method	PSNR ↑	SSIM ↑	SAM ↓	NDVI ↓	M5 ↓
Bicubic	48.013	0.9583	0.837	0.01256	0.00074
CNN L1	44.087 ±0.133	0.9145 ±0.0025	1.564 ±0.042	0.02562 ±0.00048	0.00521 ±0.00012
CNN L1+λSAM	43.956 ±0.098	0.9118 ±0.0023	1.568 ±0.026	0.02565 ±0.00083	0.00529 ±0.00009

A1: GRADIENT ALIGNMENT DURING TRAINING



Mean over the second half of training: 0.813

FINDINGS

- In-domain, R1 exceeds bicubic by +1.61 dB PSNR and improves both spectral metrics, which confirms that the implementation is correct.
- On the held-out site, R1 falls 3.93 dB below bicubic. The training sites are European coastal and agricultural terrain; FGMANAUS is tropical rainforest, where the blue and red bands are four to eight times darker than the training range. The model applies a positive bias of approximately 0.003 reflectance to the visible bands, a relative error near 75% on red, which propagates to NDVI.
- R2 is not distinguishable from R1: PSNR differs by -0.13 dB against a seed standard deviation of 0.13 dB. Mean gradient alignment is 0.81. The SAM metric does not improve under SAM training.